# FIN 2020, Final Exam

**February 7, 2023**

1. Consider portfolios formed from two risky assets, the first with expected return equal to $\mu_1 = 10$ and standard deviation of its return equal to $\sigma_1 = 4$ and the second with expected return equal to $\mu_2 = 7$ and standard deviation of its return equal to $\sigma_2 = 2$. Let $w_1$ denote the fraction of wealth in the portfolio allocated to asset 1 and $w_2$ the fraction of wealth allocated to asset 2.

   (a) (5 points) Calculate the expected return and the standard deviation of the return on the portfolio that sets $w_1 = 1/2$ and $w_2 = 1/2$, assuming that the correlation between the two returns is $_{12} = 1$.

   (b) (5 points) Calculate the expected return and the standard deviation of the return on the same portfolio that sets $w_1 = 1/2$ and $w_2 = 1/2$, assuming instead that the correlation between the two returns is $_{12} = 0$.

   (c) (5 points) Finally, suppose that in addition to the two risky assets described above, there is also a risk-free asset with return $r_f = 5$. Still assuming, as in part (b), that the correlation between the two random returns is $_{12} = 0$, calculate the expected return and the standard deviation of the return on the portfolio that allocates the fraction $w_1 = 1/4$ of wealth to asset 1, $w_2 = 1/4$ of wealth to asset 2, and the remaining fraction $w_r = 1/2$ to the risk-free asset.

2. True or False or "it depends". Briefly explain or qualify your answer.

(1) (4 points) Diversification works only when asset returns are uncorrelated.

(2) (4 points) CAPM says that all risky assets must have positive risk premium.

(3) (4 points) CAPM says the expected return on an investment with a beta of 2 is twice as high as the expected return on the market.

(4) (4 points) If a stock lies below the security market line (SML), it is under valued.

(5) (4 points) CAPM assumes that all investors have the same information. CAPM also says they are willing to hold the market portfolio.

3. Consider the following estimates of market model for two stocks:

$$\tilde{r}_1 = -0.002 + 0.81\tilde{r}_M + \tilde{\varepsilon}_1,$$
$$\tilde{r}_2 = 0.004 + 1.5\tilde{r}_M + \tilde{\varepsilon}_2,$$

where the standard deviation of $\tilde{\varepsilon}_1$ and $\tilde{\varepsilon}_2$ are $\sigma_{\varepsilon_1} = 0.06$ and $\sigma_{\varepsilon_2} = 0.13$ respectively. The market return has a mean of 8% and standard deviation of 17%, and risk-free rate is 2%.

(a) (5 points) Find the covariance matrix of stock returns.

(b) (5 points) Consider an equal-weighted portfolio $p$ of the stocks. Is it a well-diversified portfolio? Calculate the beta and expected return of the portfolio.

(c) (5 points) What is the total risk of the portfolio? Is it entirely compensated by the market?

(d) (5 points) Create an efficient portfolio $q$ whose total risk is equal to the aggregate risk of $p$. What is the expected return of $q$? Explain your result.

$$E(\tilde{r}_p) = r_f + \beta(E(\tilde{r}_m) - r_f)$$

4. Consider a two-date economy where there are three states of the world at date 1. There are two risky assets and a risk-free asset. The payoff of the risky stock at date 1 will be 2, 4, or 6, and it trades at the price of 3 at date 0. The payoff of the second risky stock at date 1 will be 3, 5, or 7, and it trades at the price of 5 at date 0. The risk-free asset pays a net return of 2%. The probabilities of the three states are 0.3, 0.5, and 0.2, respectively.

   (a) (5 points) Are markets complete? If not, propose a solution to complete it.

   (b) (5 points) Consider the original states. Eliminate the second risky asset and replace it with a call option of the first risky asset with a strike price of 4. Are markets complete now?

   (c) (5 points) Compute the Arrow-Debreu prices and interpret the results.

   (d) (5 points) Price a put option on the first risky stock with an exercise price of 5 by using the Arrow-Debru prices.

   (e) (5 points) Calculate the stochastic discount factor and the risk-neutral probability, and price again the put option in two other ways different from the one used in part (d).

$P_0 = 3$ $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ $P_0 = 1$ $\begin{pmatrix} 1.02 \\ 1.02 \\ 1.02 \end{pmatrix}$ $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$

$\frac{5}{12} + \frac{2}{12} + \frac{3}{12} = \frac{10}{12}$

5. Consider a two-period exchange economy with 2 consumers. Consumer $j$'s utility is

$$U_j = \ln c_j^0 + \mathrm{E}[\ln \tilde{c}_j]$$

where $j = 1, 2$, $c_i^0$ is consumer $j$'s date-0 consumption, and $\tilde{c}_j$ is consumer $j$'s date-1 consumption which depends on the state $i$. There are two states of nature at date 1, i.e, $i = 1, 2$. Let's use $w_j^0$ denote consumer $j$'s date-0 endowment, and $w_j^i$ denote consumer $j$'s date-1 endowment when state $i$ occurs. The two consumers have the following endowments: $w_1^0 = w_2^0 = 4$, $w_1^1 = w_2^2 = 1$, and $w_1^2 = w_2^1 = 3$.

(a) (5 points) Describe a Pareto-optimal allocation of resources. Is it unique?

(b) (5 points) Suppose only one type of security may be issued. Is it possible to achieve the Pareto-optimal allocation? If so, how?

(c) (5 points) Suppose the consumption good can be costlessly stored. No other asset is traded. Describe intuitively how this opportunity will be exploited and why. Is the utility level of the two consumers increased?

(d) (5 points) Now assume consumer 1 is risk neutral, and there is aggregate uncertainty, that is, at date 1 the endowments are $w_1^1 = 1$, $w_1^2 = 2$, $w_2^1 = 3$, and $w_2^2 = 4$. Discuss intuitively what the new Pareto-optimal allocation should look like. Assume the existence of Arrow-Debreu securities and characterize the resutling market equilibrium.